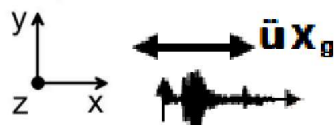
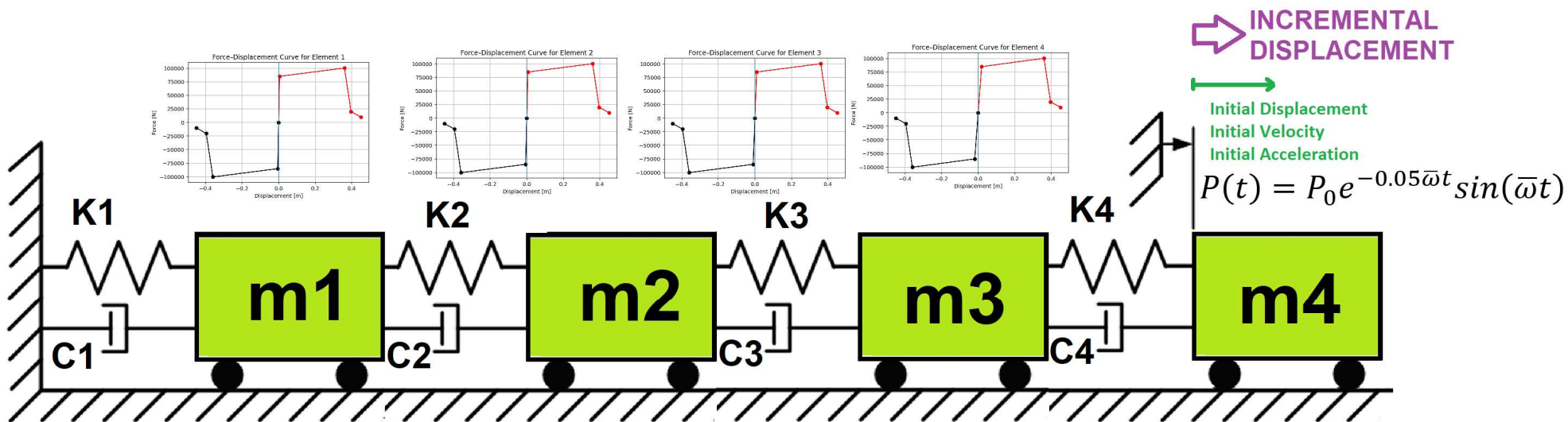


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE EVALUATION OF THE EQUIVALENT VISCOUS DAMPING RATIO OF A NONLINEAR MDOF SYSTEM UNDER MULTIPLE LOADING PROTOCOLS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

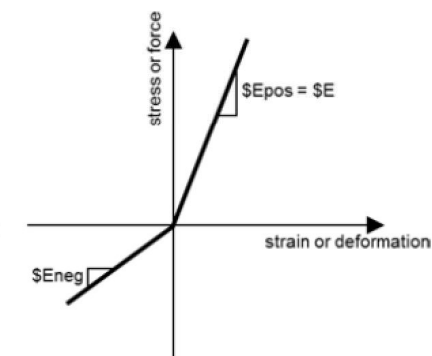
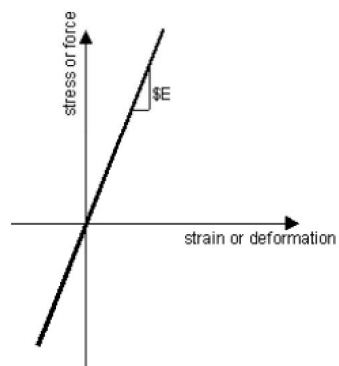
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

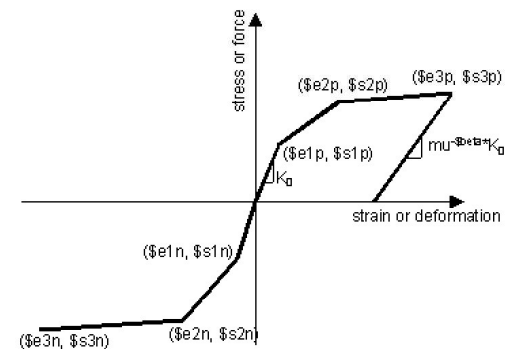
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

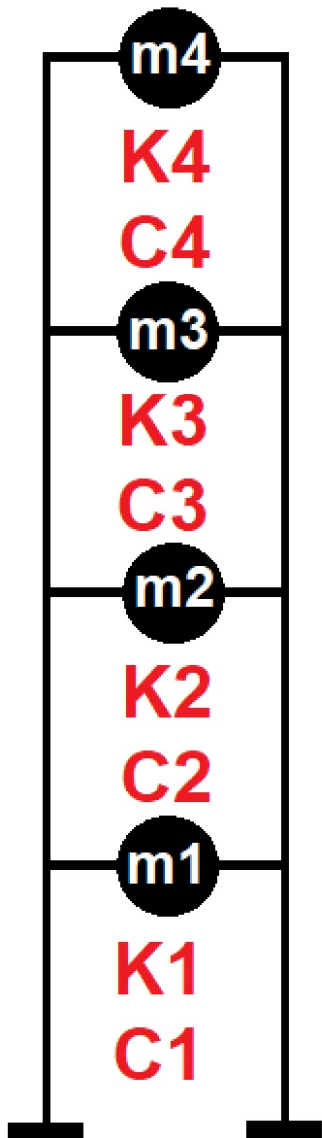
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



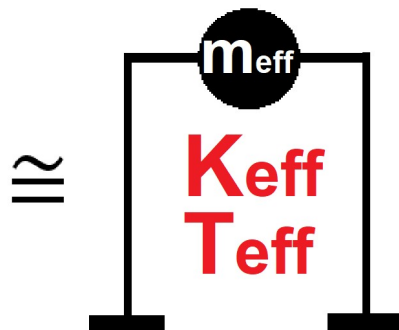
ELASTIC MATERIAL



INELASTIC MATERIAL

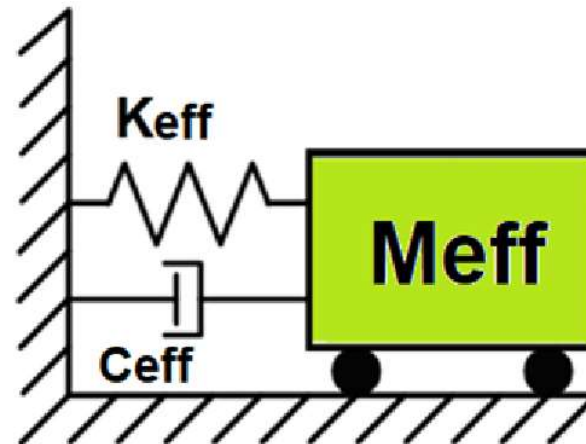


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

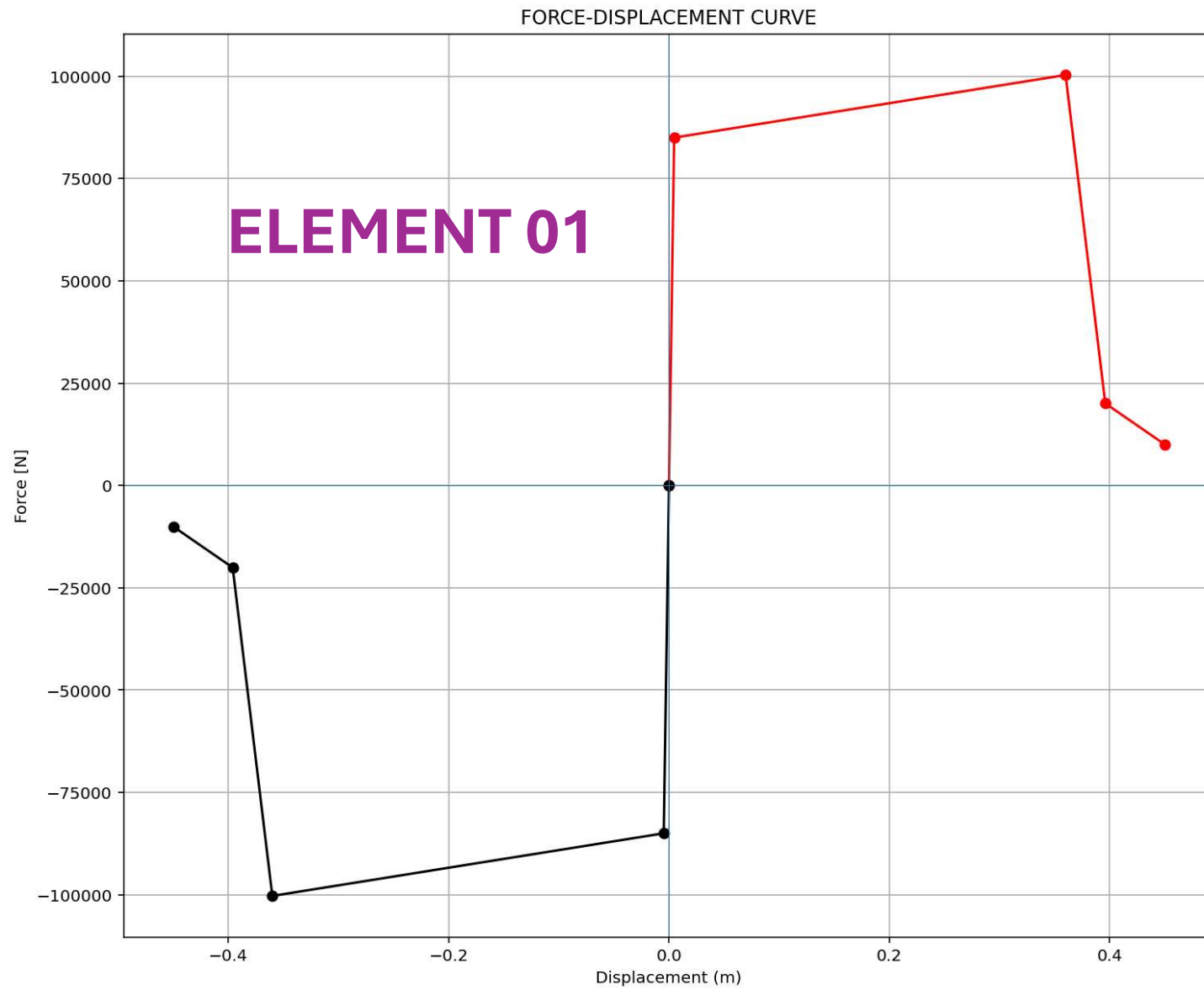
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

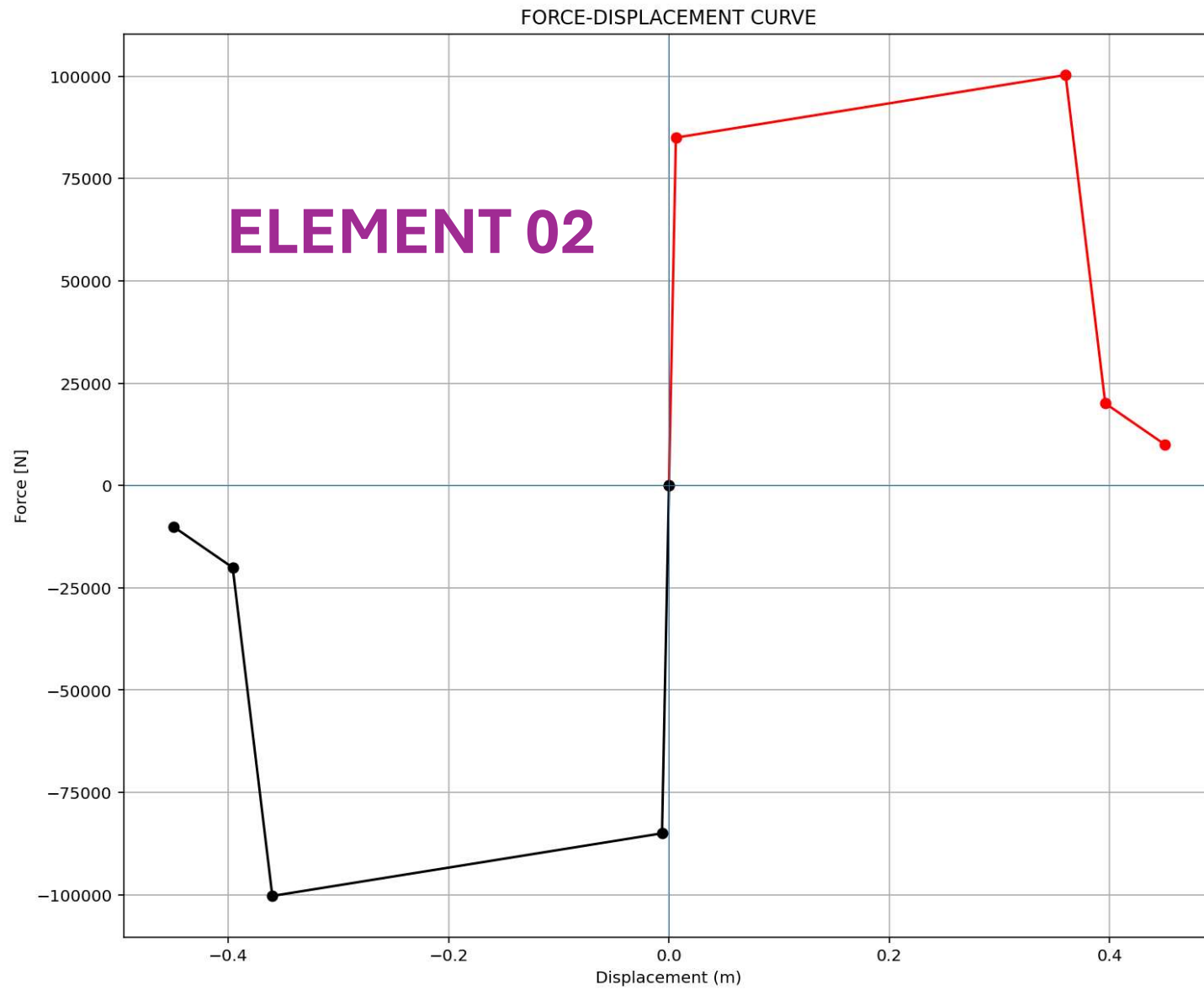
where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{\text{sec}} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

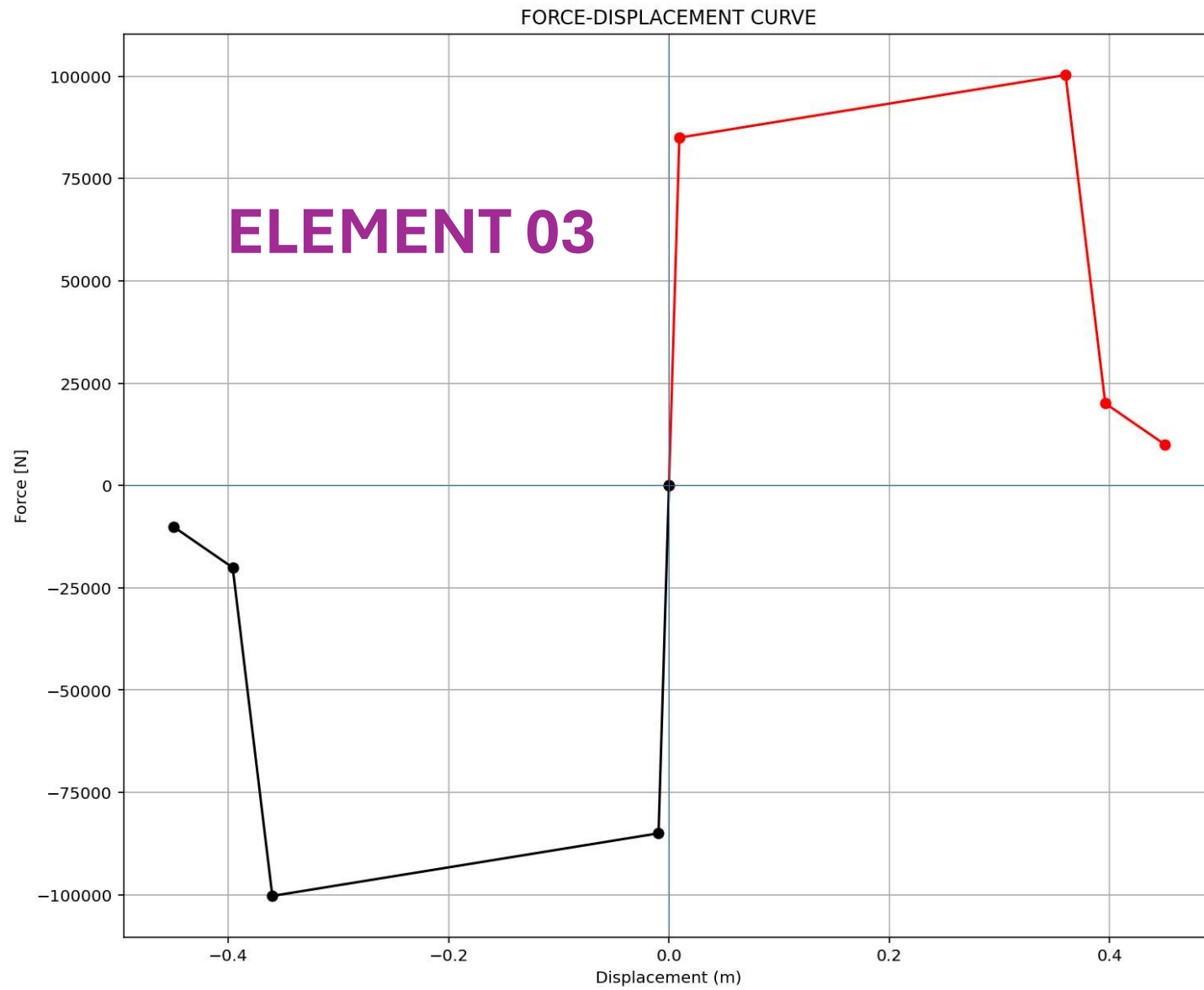
The equivalent viscous damping ratio is then:

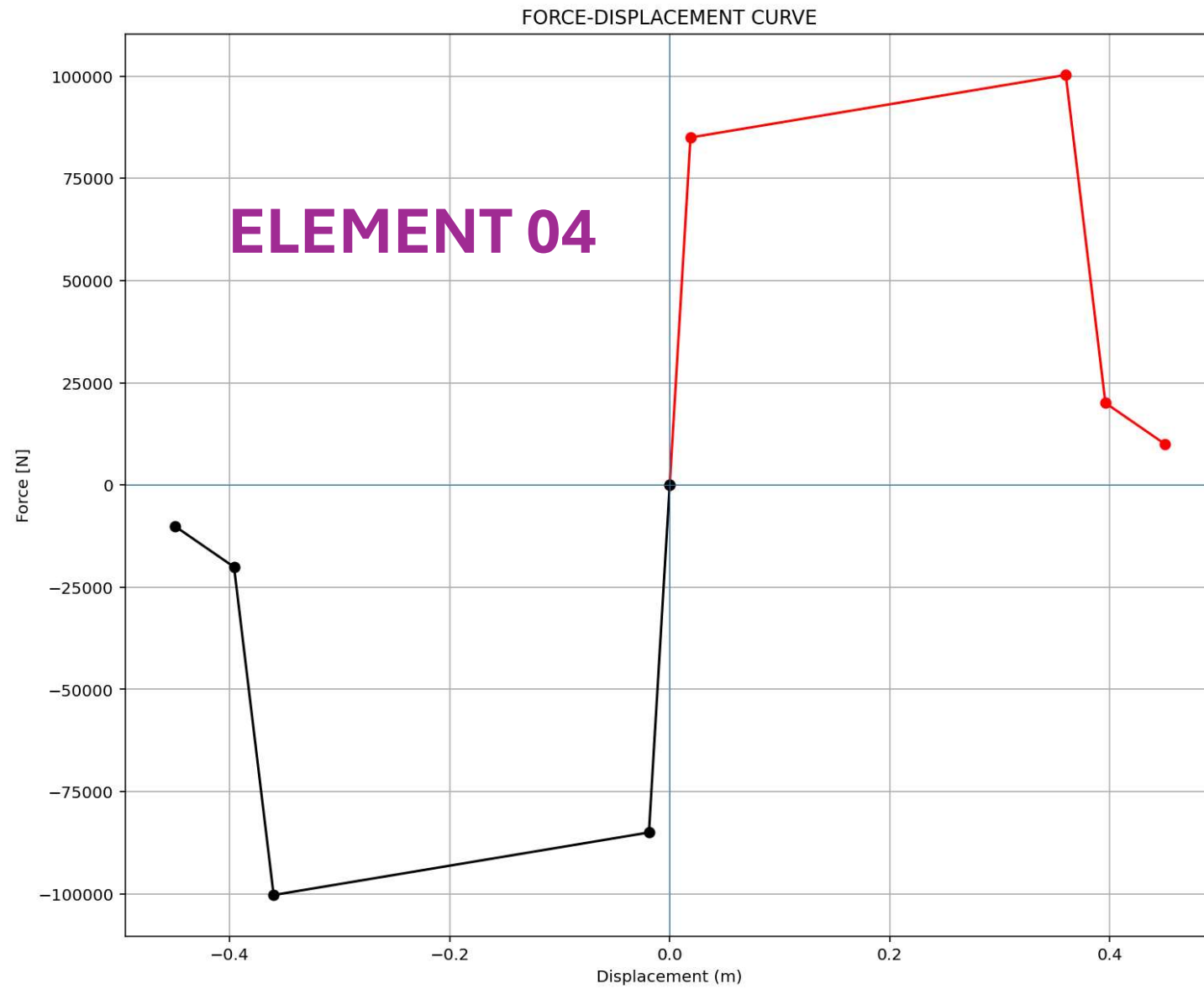
$$\xi_{eq} = \frac{100 E_d}{4\pi E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.









FileEditSearchSourceRunDebugConsolesProjectsToolsViewHelp

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...N_CAPACITY_INDEX\22_MDOF_8_ANA\P(t)_MDOF_8_ANA.py

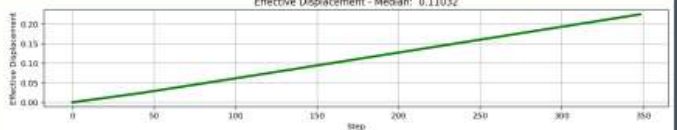
P(t)_MDOF_8_ANA.py X

COMPUTE_EFFECTIVE_PROPERTIES_FUN.py X

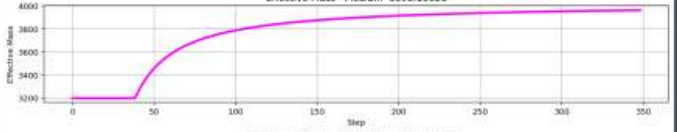
```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM STRUCTURE : AN OPEN
4 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
5 #-----
6 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANAL
7 #-----
8 #
9 # P(t) = P0 sin(wt)
10 # P(t) = P0 exp(-0.05wt) sin(wt)
11 #-----
12 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
13 # OF ENERGY DISSIPATION CAPACITY INDEX
14 #-----
15 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
16 # EMAIL: salar.d.ghashghaei@gmail.com
17 #####
18 Nonlinear Seismic Performance Assessment of a MDOF:
19 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
20
21 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
22 for performance-based earthquake engineering. The 2D model incorporates both material non
23 (elastic-perfectly plastic or hysteretic steel with strain hardening)
24 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
25 and large displacements-essential for collapse assessment.
26
27 Eight analysis protocols are implemented:
28 (1) [PERIOD] : Structural Period
29 (2) [STATIC] : Gravity load analysis establishing dead load state
30 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
31 and plastic mechanism identification
32 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
33 pinching behavior, and energy dissipation degradation
34 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
```

26 %

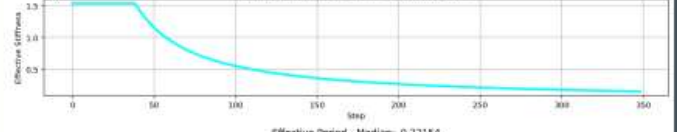
Effective Displacement - Median: 0.11032



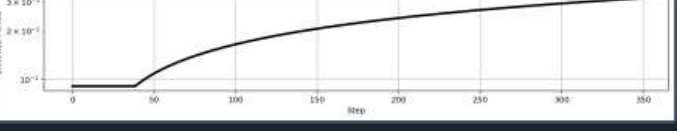
Effective Mass - Median: 3898.10026



Effective Stiffness - Median: 3135556.42315



Effective Period - Median: 0.22154



IPython Console

Files

Help

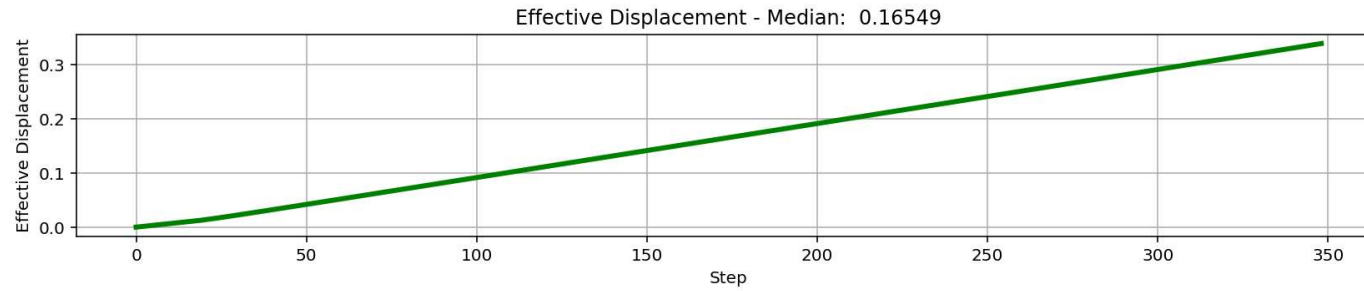
Variable Explorer

Debugger

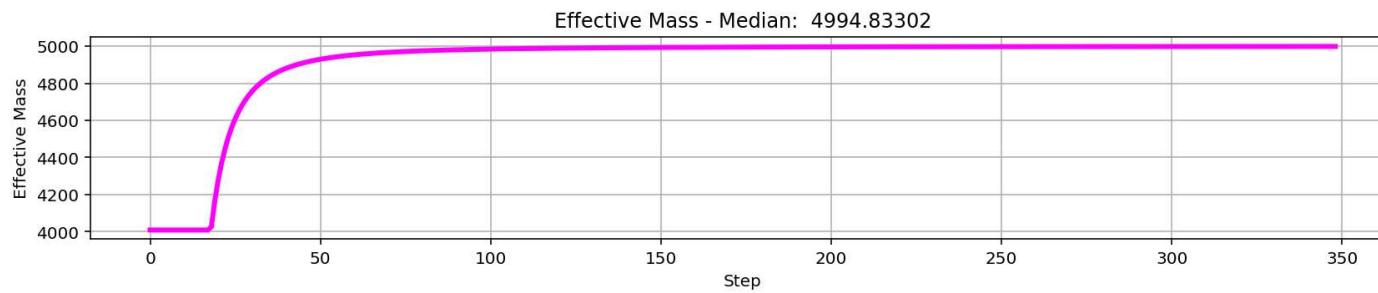
Plots

History

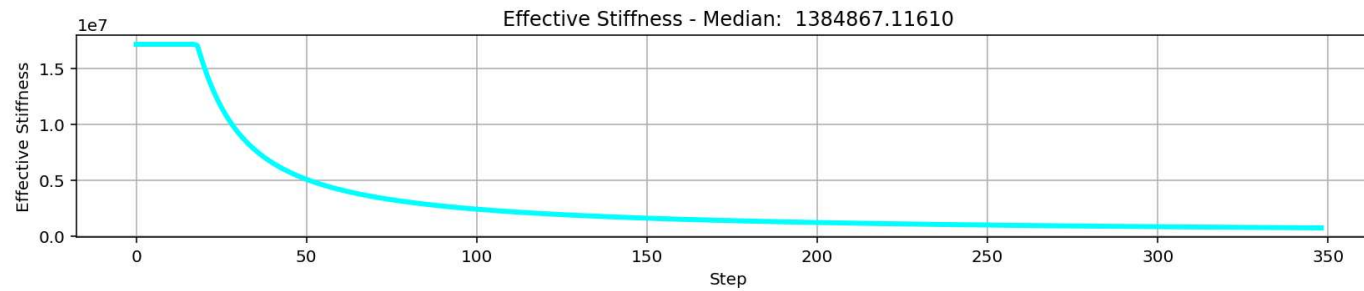
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 6, Col 108 UTF-8 CRLF RW Mem 48%



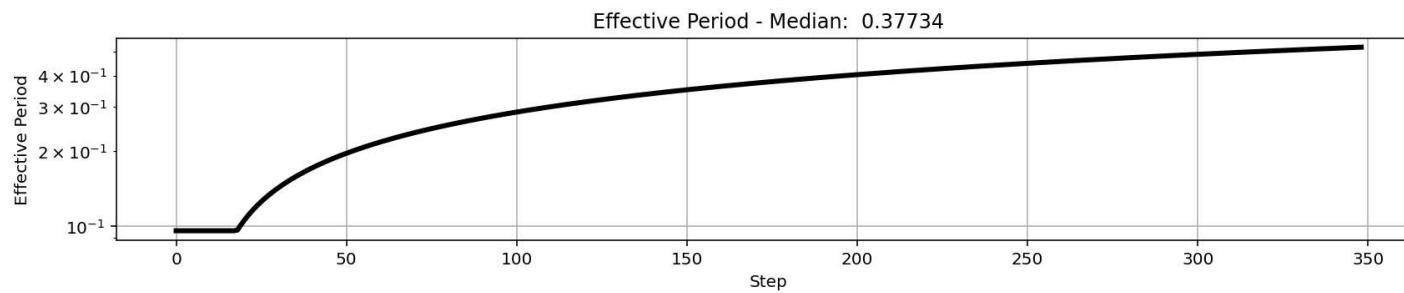
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



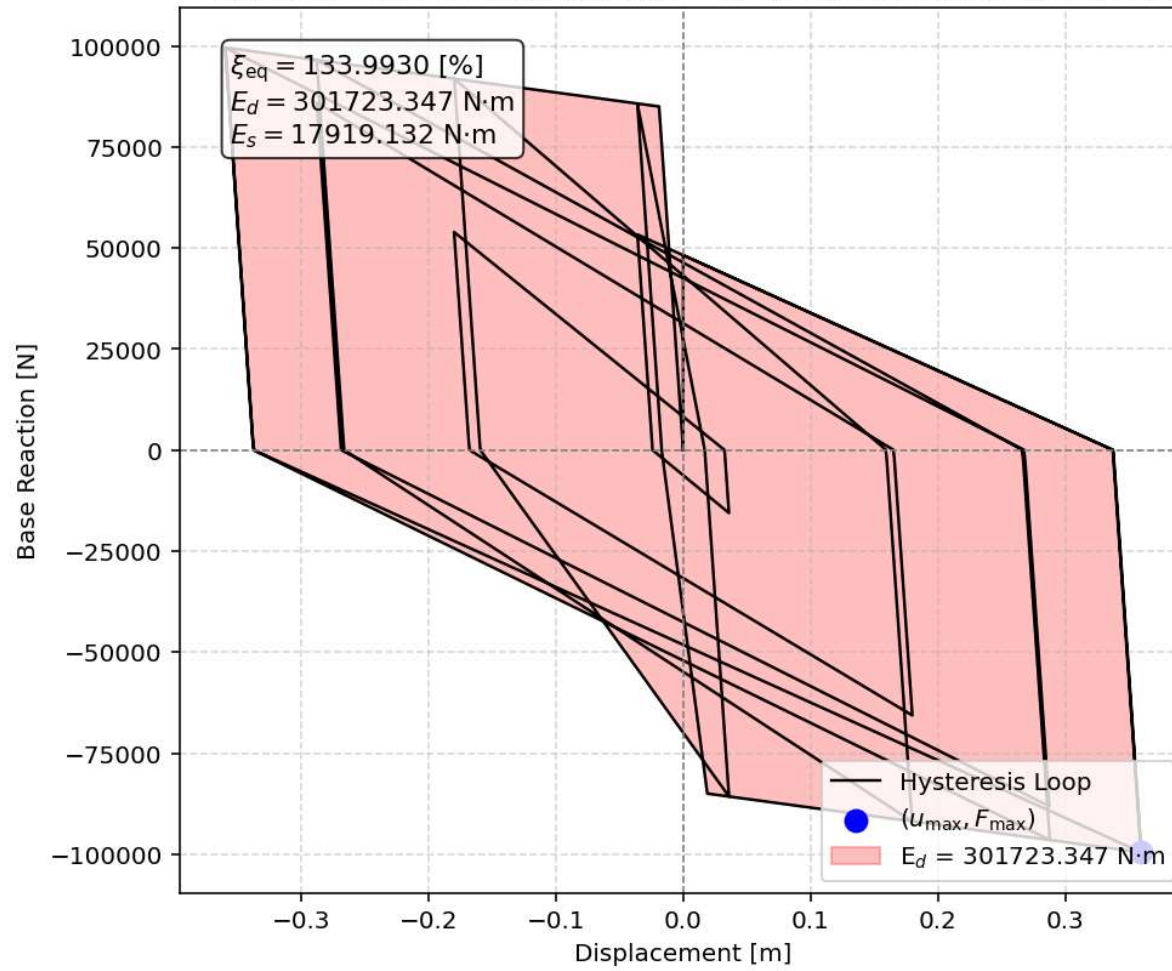
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$



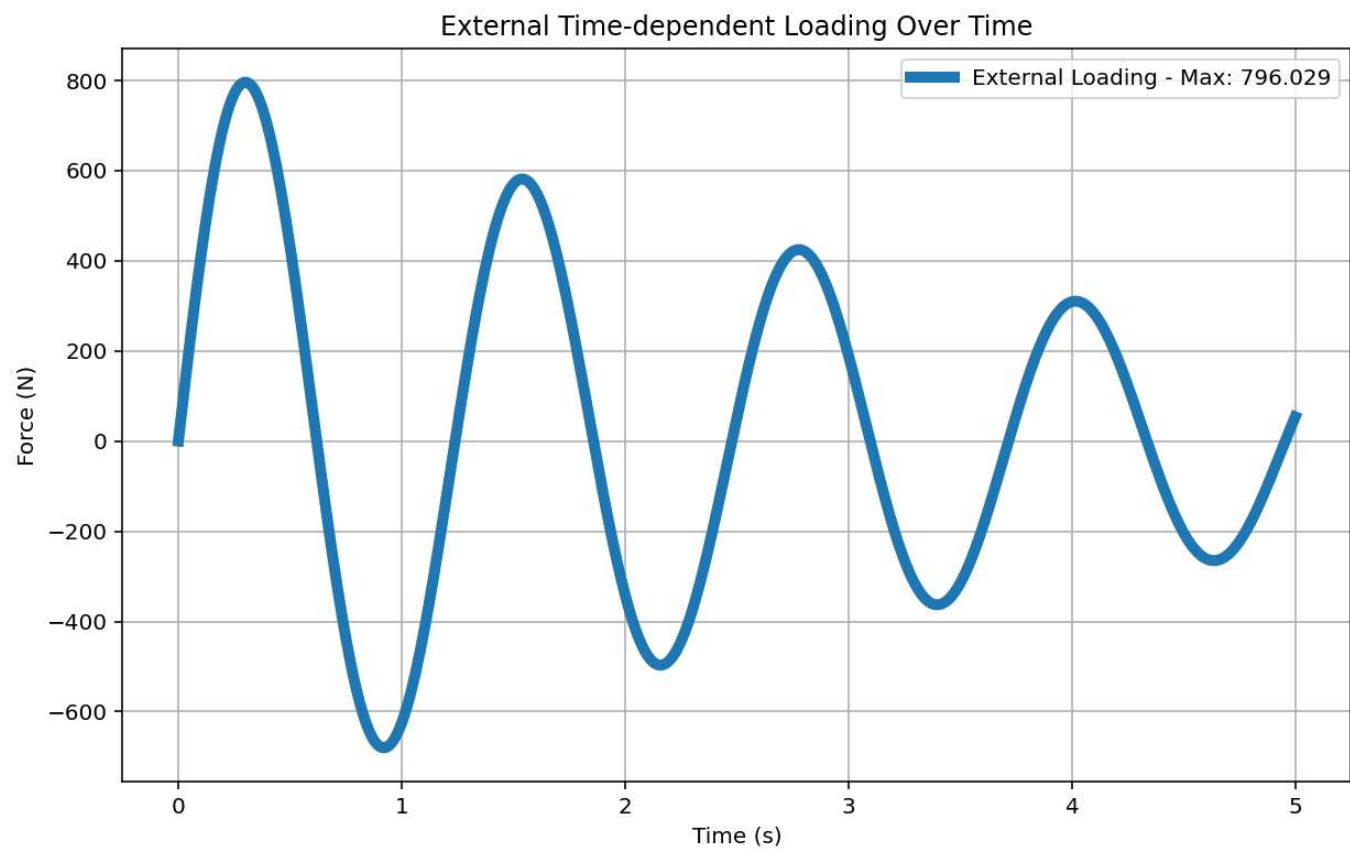
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS

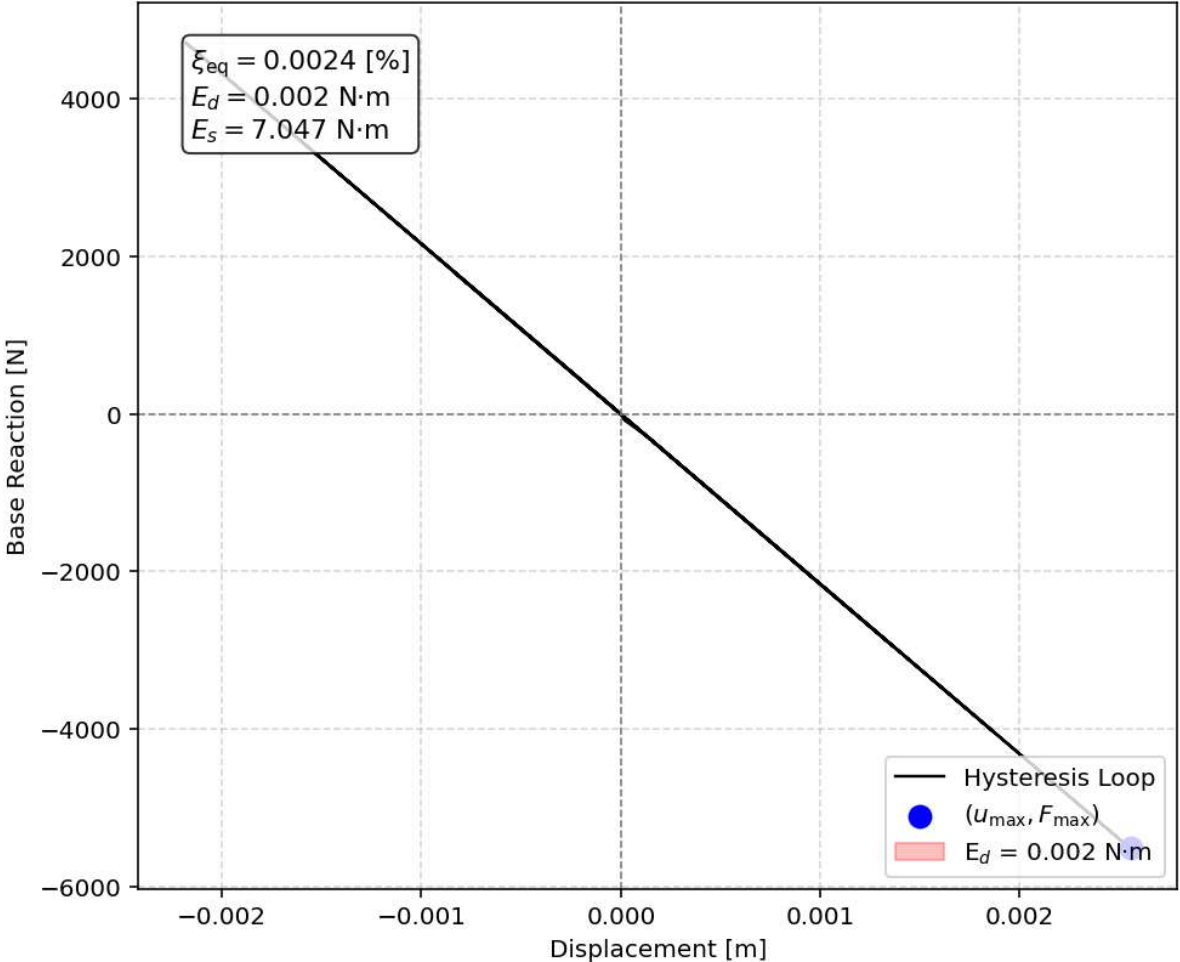
Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

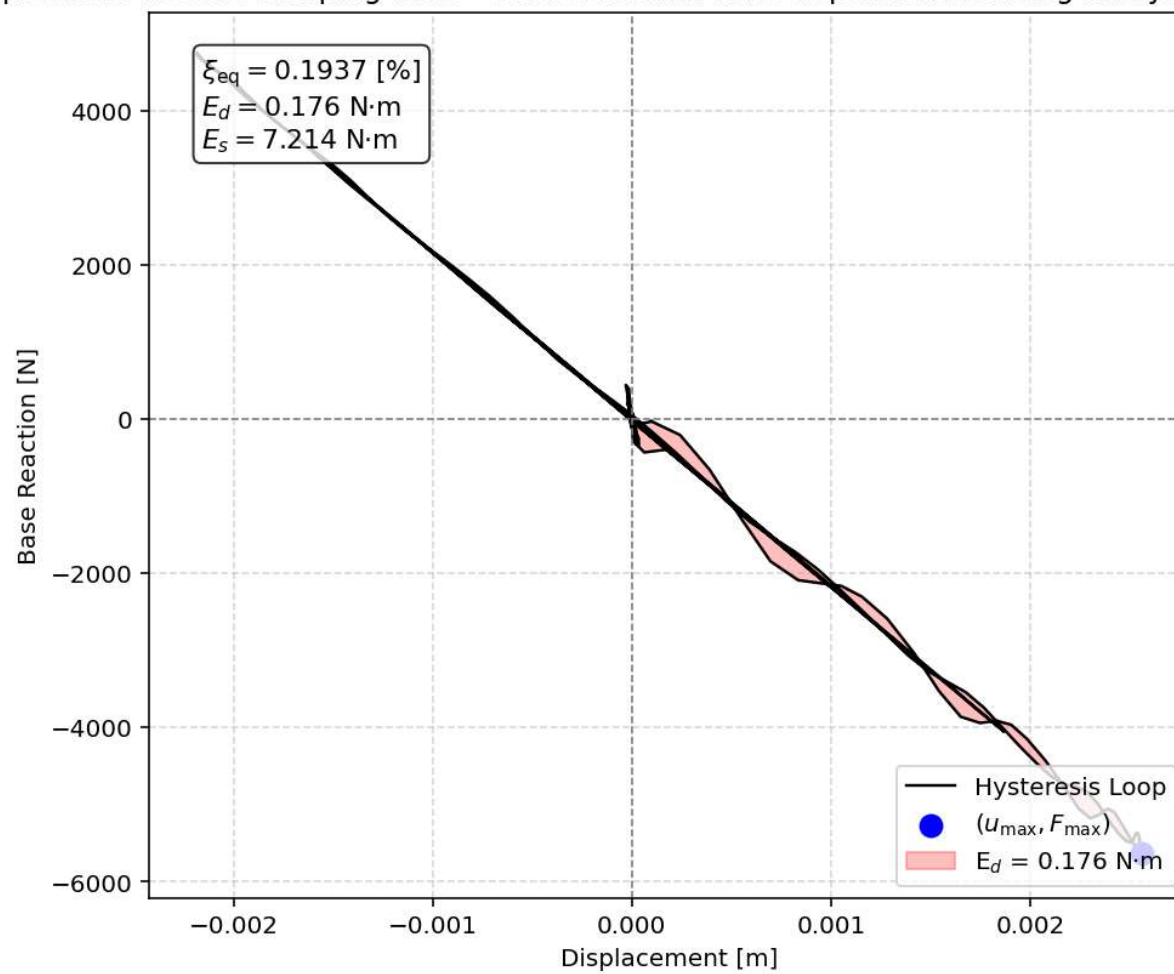


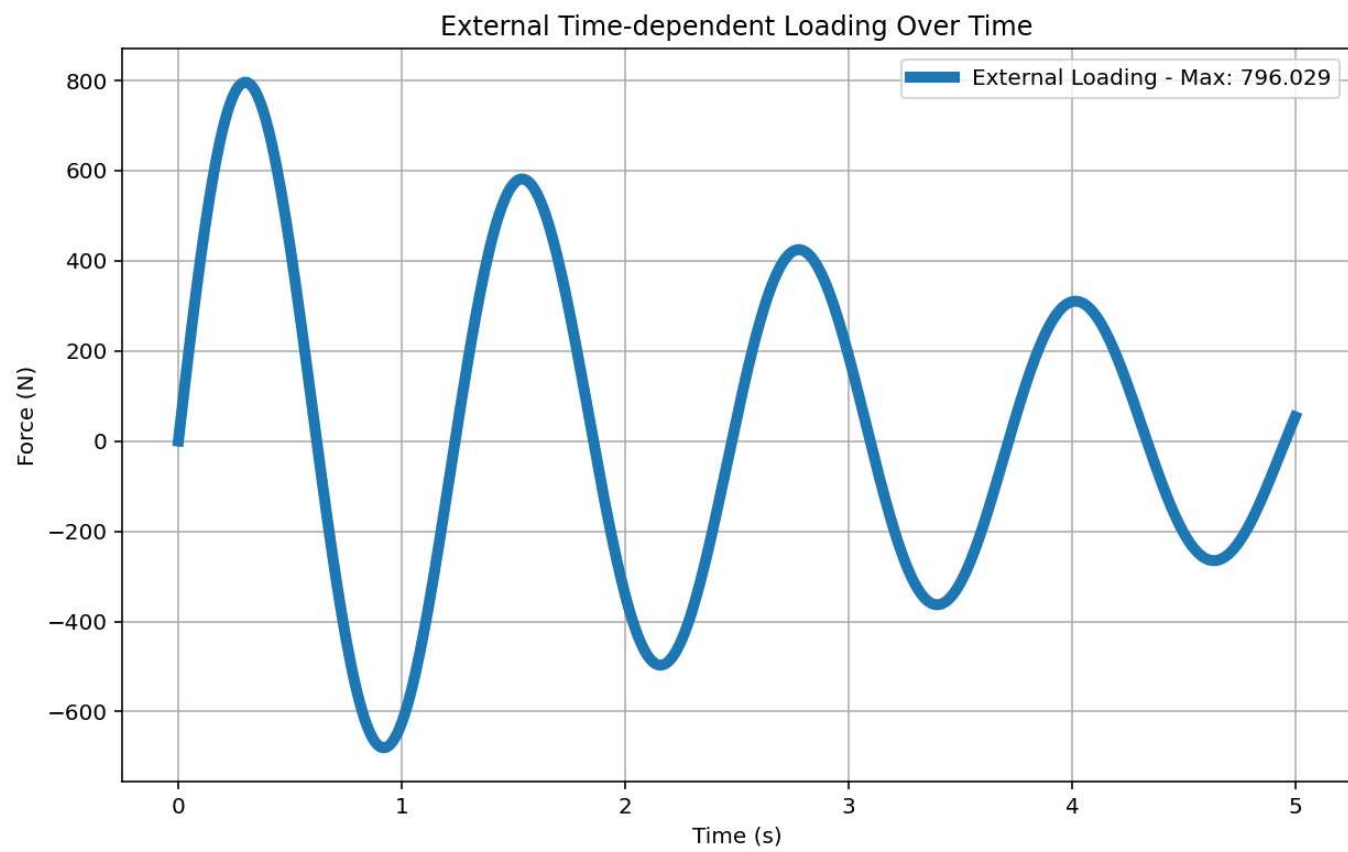
Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis



DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis





FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...OMPUTE_EFFECTIVE_PROPERTIES_FUN_FREE_VIBRATION.py

P(t)_MDOF_8_ANA.py COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py

```
1  """ EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH
2  # Change MDOF to SDOF System with Displacement Based Design Concept
3  # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
4  """
5  This script implements a displacement-based pushover transformation,
6  converting a multi-degree-of-freedom (MDOF) system into an equivalent
7  single-degree-of-freedom (SDOF) system for seismic assessment.
8  It calculates effective modal properties-displacement, mass, and
9  stiffness-by weighting element forces and nodal displacements according
10 to a presumed deformed shape.
11 The derived effective period provides a simplified dynamic characteristic
12 for performance-based engineering. The visualizations effectively track the
13 evolution of these equivalent parameters throughout the nonlinear analysis steps.
14 """
15
16 #-----
17 import numpy as np
18
19 displacement_X_1 = np.array(list(node_displacements.values())[1]) # DOF 02
20 displacement_X_2 = np.array(list(node_displacements.values())[2]) # DOF 03
21 displacement_X_3 = np.array(list(node_displacements.values())[3]) # DOF 04
22 displacement_X_4 = np.array(list(node_displacements.values())[4]) # DOF 05
23
24 ele_force_01 = np.array(list(ele_force.values())[0]) # ELEMENT 01
25 ele_force_02 = np.array(list(ele_force.values())[1]) # ELEMENT 02
26 ele_force_03 = np.array(list(ele_force.values())[2]) # ELEMENT 03
27 ele_force_04 = np.array(list(ele_force.values())[3]) # ELEMENT 04
28
29 STIFF_X_1 = np.abs(ele_force_01 / displacement_X_1)
30 STIFF_X_2 = np.abs(ele_force_02 / displacement_X_2)
31 STIFF_X_3 = np.abs(ele_force_03 / displacement_X_3)
32 STIFF_X_4 = np.abs(ele_force_04 / displacement_X_4)
33
34 MX2 = (MASS[0] * np.square(displacement_X_1) +
        MASS[1] * np.square(displacement_X_2) +
```

Effective Displacement - Median: 0.00002

Effective Mass - Median: 40.05 (5406)

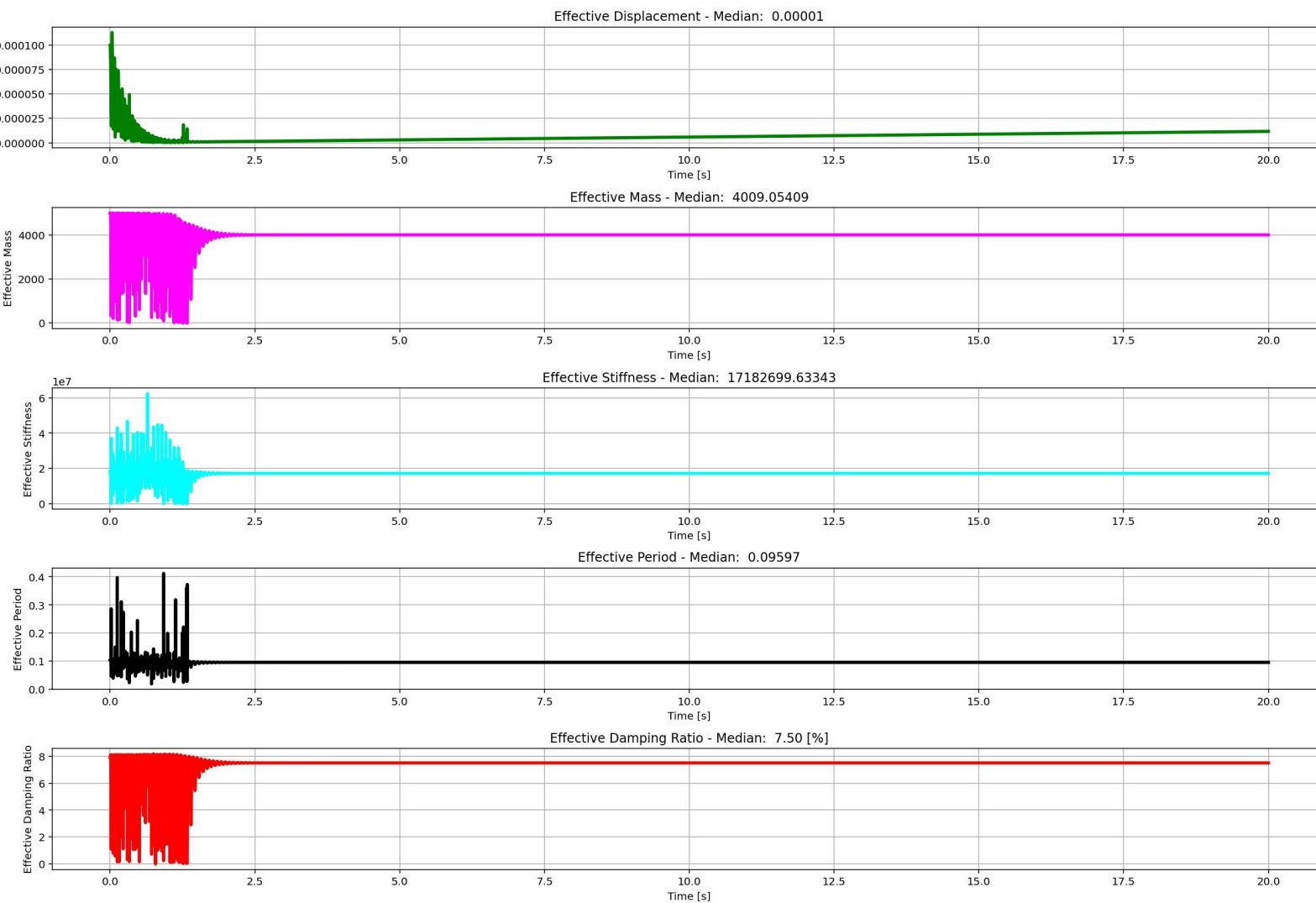
Effective Stiffness - Median: 37182099.63361

Effective Period - Median: 0.00597

Effective Damping Ratio - Median: 1.54 (%)

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 7, Col 55 UTF-8 CRLF RW Mem 45%



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$

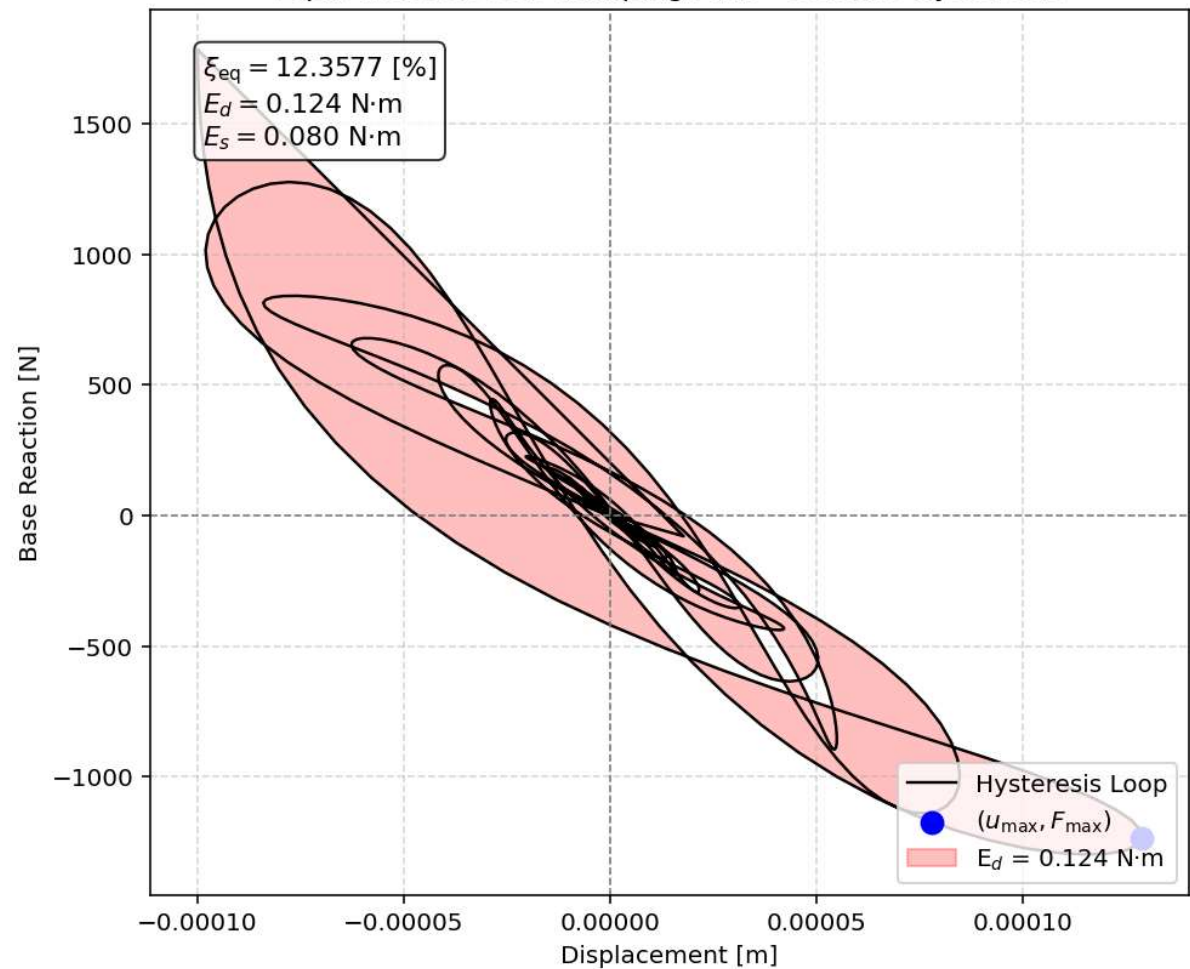
$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$

$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

$$\xi_{eff} = \frac{\sum_{i=1}^n \xi_i \Delta_i}{\Delta_d}$$

Equivalent viscous damping ratio - Seismic Hysteresis



SEISMIC ANALYSIS

Equivalent viscous damping ratio - Seismic Hysteresis

